STATEMENT OF AD HOC COMMITTEE ON PROJECT INDEPENDENCE

(This is a statement of the undersigned and not of organizations with which they are affiliated.)

As we understand it, **Project Independence** is aimed at making the United States substantially independent of foreign sources of energy or at least to acquire the capability to do so. **We support this goal**, and we are concerned that there has not yet been developed a national policy that correctly identifies short term and long term problems and correctly assigns roles to the public and private sectors in solving them.

The U.S. Energy Position

In the long term, the U.S. and the world have good prospects for energy. Either nuclear or solar energy by itself can provide enough energy so that the population the earth can support will be limited by something else.

The U.S. is in a particularly good long term position, because of its resource base, its established light and heavy industry, and its technological manpower. There is no moral imperative to reduce our per capita energy consumption, *provided we contrive to obtain our energy from within our own borders*. Moreover, we probably can give other countries some help in solving their energy problems.

The U.S. has serious short term problems stemming from the decreasing availability of domestic petroleum and profiteering by the petroleum exporting countries. Other countries, especially the underdeveloped countries, have even more serious and more immediate problems.

Environment and the Demand for Energy

Energy use and clean air and water are both essential components of the standard of living in this country. The large *per capita* use of energy has made possible a substantial part of our individual freedom and prosperity, but our health and our enjoyment of our freedom and prosperity are reduced by environmental neglect some of which comes from the ways energy has been obtained. Fortunately, presently known technology permits both further increases in our use of energy and further improvement of the environment. In the short run, the rate at which we improve the environment and the rate at which we increase the standard of living of the various groups in our population compete. The political process must mediate this competition as it mediates all competitions for public resources.

Prediction of future energy demand by extrapolating exponential growth curves is unrealistic. The growth in American cattle production between 1860 and 1880 would have led to predictions that each American would have to eat a cow a day long before now. Our ability to use energy *for its present applications* will saturate in the near future, probably before the year 2000. Thus registered automobiles have grown from 30 million to 100 million since World War II; they cannot grow by that ratio again in the next thirty years without there being more cars than people. On the other hand, home air-conditioning has just started its fast growth phase in many parts of the country.

In the long run, there is no need to distinguish energy from other commodities or from human labor by a special conservation ethic. Energy is a commodity like any other, and if its price expresses all the costs that go into its production, people will regulate their own use of it in accordance with their own interests better than the government or preaching can do. In the short run, emergencies calling for special conservation measures can occur, and a special effort to call peoples's attention to ways in which energy is wasted are worthwhile. However, there is some tendency to use the energy crisis to scare people into making changes in life style that some intellectuals consider desirable for other reasons. Officials should resist any temptation to do this, because it is dishonest, and when the dishonesty is discovered, public distrust will be increased.

Goals for Project Independence

Since petroleum and natural gas are the high priced commodities in short supply, since U.S. production of them has probably hit its peak, and since they will be all gone in the forseeable future, many of the goals of **Project Independence** can be put in terms of reducing our dependence on them.

1. In the short run, we can increase the domestic supply by finding more oil fields especially in the offshore areas that have not been explored yet. We can also further employ secondary recovery methods.

2. We can also reduce demand by eliminating waste. If necessary, we can make people use smaller cars and use public transportation. However, these measures are not necessary in the long run, and the resulting infringement on individual freedom should be avoided if possible and limited in time if not.

3. It should be an early goal of **Project Independence** to eliminate the use of natural gas and petroleum for the central station generation of electric power. Already coal and nuclear energy are cheaper and the price differential will grow with time. Some of the vagueness of **Project Independence** will vanish once we can set a target date for this conversion.

4. Next we should provide substitutes for natural gas and natural petroleum for space heating. It can be done by a combination of solar heating, heat pumps, synthetic gas made from coal and central station hot water. This step will take longer than elimination of petroleum use to produce electricity and makes sense only after the former has been accomplished, since it is more efficient to use petroleum directly for space heating than to use electricity generated by petroleum for that purpose.

5. It will take still longer to provide substitutes for petroleum for use in vehicles. Synthetic gasoline from coal is the most immediate possibility with hydrogen a longer term bet. Electric vehicles seem attractive, but it is not yet clear that a suitable battery is possible.

Research and Development

All potentially cost-effective sources of energy or ways of conserving it should be explored. This includes funding parallel approaches to important problems. It is important to avoid the kind of wishful thinking that eliminates one source of energy because another would be better when it is not yet clear that the latter can be developed in a reasonable time to provide energy at a reasonable cost. People who want to develop one approach often have a tendency to knock potential rivals.

We think the following approaches are important:

1. We can find new methods of exploring for oil and gas and new secondary recovery techniques. It should be clearly understood that this only buys us time. We should try to reduce our dependence on foreign oil faster than we are forced to, because other countries cannot reduce their dependence as fast as we can. In other words, we should stop outbidding the underdeveloped countries for oil as soon as possible.

2. Synthetic petroleum from oil shale and from coal all should be developed. Which will win depends on economic and environmental considerations.

3. The present breeder reactor projects should be completed and alternate breeder approaches investigated. The improvement of non-breeder reactors should be continued.

4. We should expand the research on fusion power, but until scientific feasibility is demonstrated, we cannot count on it for a specific date.

5. Solar energy should be pushed for those applications for which it may be cost-effective like space heating. Central station solar power should be explored, but again we can't count on it for a specific date until a potentially cost-effective way of producing it is developed.

6. Some of the potential ways of using energy more economically such as better automobile engines and transmissions deserve extensive research as does building insulation.

7. Research in exotic sources of energy, e.g. geothermal, wind tide, and oceanic temperature difference should be supported whenever an objective investigation shows a reasonable chance of cost-effectiveness. There is a tendency to believe that unexplored methods won't have the unpleasant side-effects of known methods, and there is a tendency to continue wishful thinking by making "demonstration projects" that don't lead to any plausible path to cost-effectiveness.

Nuclear Energy

Nuclear energy plays a special role in public thinking because of the controversy about its safety.

There are questions about the safety of reactors, emissions from reactors, the storage and handling of radioactive wastes and about the possibility of theft of nuclear materials by terrorists. New dangers and new ways of meeting them continue being discovered, and we will never attain absolute safety or even absolute knowledge about the degree of danger that currently exists. As scientists and engineers, we would especially like to have more information.

However, decisions have to be made now, and our considered judgment is that the nuclear energy program should proceed without additional general delays or moratoriums, and should even be speeded up if this is needed to meet the goals of **Project Independence**. This judgment applies in particular to the current procedures for approving the construction of reactors, to the demonstration fast breeder, and to the recycling of plutonium.

We believe that present American procedures are adequate though subject to further improvement.

Some of the techniques of safety analysis developed in connection with nuclear energy should be extended to other energy systems and to other industrial activities.

In the meantime, independent of the nuclear controversy although exacerbated by it, is the fact that the nuclear industry is faltering badly and cannot meet the commitments it has already made without a concerted effort of government and industry. The capacity of several components of the fuel cycle is likely to prove less than required by the projected power plants. These clearly include uranium enrichment and spent fuel reprocessing and probably also include spent fuel storage and waste disposal.

The International Energy Picture

Never in history has the price of such an important commodity of international commerce been raised so far above costs of production by a cartel. Never have the effects on so many countries been so serious. The United States, of course, is one of the lesser sufferers, because we still produce most of our own energy and because we can probably afford the exorbitant prices.

The other industrial countries and the underdeveloped countries have even more immediate energy problems than we do. Therefore, they are more likely to have to undertake crash programs than we are. Unless we want to suffer the ill consequences of their failure, we should help insure their success by helping develop technology. In particular, the pace of development of nuclear energy, automation of coal mining, coal gasification, and oil shale should be partly determined by the needs of others.

In addition, we can probably get in a position to export energy. We should continue exporting coal, nuclear power plants, and nuclear fuel.

Crash Programs

If the international price of oil can be talked down, if there is substantial increased investment in present energy sources, and if there is a prompt start on developing replacements for petroleum, then crash programs are probably unnecessary - at least in the U.S.

However, the experience of World War II tells us that when necessary we can greatly increase our progress if we are willing to sacrifice some domestic consumption temporarily and also to make decisions promptly. For example, the Hanford plutonium production reactors handle about the same power as a modern nuclear power plant. It was in operation two years after the first chain reaction, but now a nuclear power plant takes ten years. It is said that we can't expand our nuclear program because of a shortage of nuclear engineers, but when the Hanford plant was started, there were no nuclear engineers at all. The same considerations apply to possible crash programs in other energy areas, but in the nuclear field, we can directly compare performance under routine and emergency conditions.

Once specific goals are established, FEA should promptly determine whether a crash program is needed to meet them. Alternatively, in determining the options to put before the President and Congress, crash programs should definitely be included.

Roles of the Private and Public Sectors

The production of energy has traditionally been a matter for the private sector, and the role of the government has been to control the rates set by natural monopolies. In addition the government has traditionally made rules to protect the environment and workers. In general, this division of function will continue, but the government has some new roles:

1. In the international sphere, the government has to bargain for the country as a whole. Private companies, especially when restrained by anti-trust from co-ordinating their policies are too weak for this.

2. Some investments may have to be made whose payoff will depend on future government policy. This includes investment in technologies that may later be unprofitable because of new environmental restrictions or because the country has done better in foreign buying than the worst case provided for. In these cases, the government has to take the risks if they are to be taken at all.

3. The government needs to co-ordinate research and development and support the energy research done in universities and non-profit institutes.

4. Finally, the government needs to stabilize the set of energy-related regulations as far as is politically possible.

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